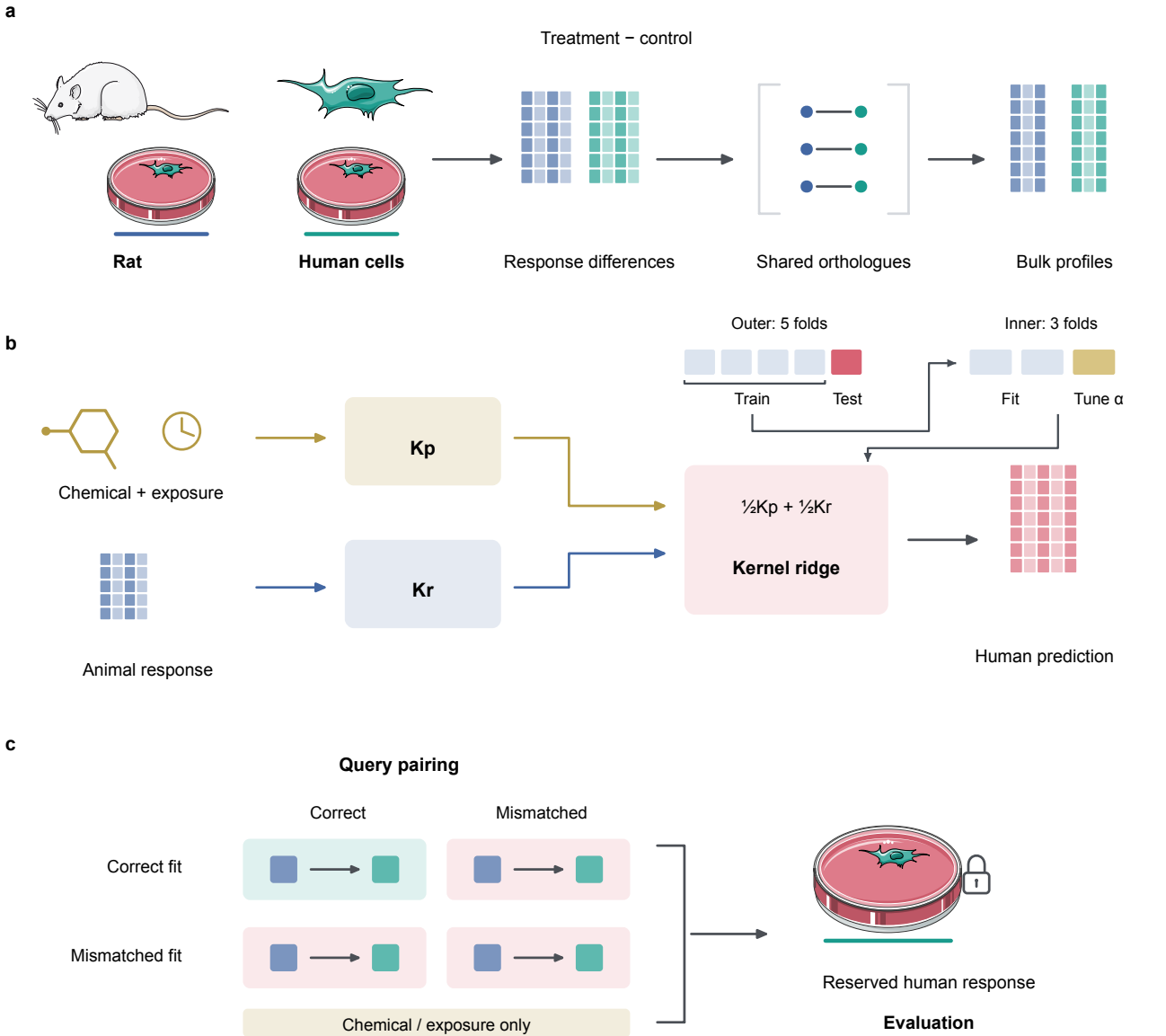

a
Treatment − control
Rat
Human cells
Response differences
Shared orthologues
Bulk profiles
b
Outer: 5 folds
Inner: 3 folds
Train
Test
Fit
Tune α
Chemical + exposure
Kp
Kr
½Kp + ½Kr
Kernel ridge
Animal response
Human prediction
c
Query pairing
Correct
Mismatched
Correct fit
Mismatched fit
Chemical / exposure only
Reserved human response
Evaluation